Developmental Psychobiology

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Developmental Psychobiology

RESEARCH ARTICLE OPEN ACCESS

Pupillary Responses to Dynamic Negative Versus Positive Facial Expressions of Emotion in Children and Parents: Links to Depression and Anxiety

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Received: 30 December 2023 | **Revised:** 30 May 2024 | **Accepted:** 5 June 2024

Funding: The contribution of Evin Aktar during manuscript preparation was supported with an NWO VENI grant #016.Veni.195.285. The contribution of Koraly Pérez-Edgar supported by the National Institute of Mental Health (NIMH), #R01 MH109692.

Keywords: anxiety | attention | depression | emotion | parents | pupillometry

ABSTRACT

Witnessing emotional expressions in others triggers physiological arousal in humans. The current study focused on pupil responses to emotional expressions in a community sample as a physiological index of arousal and attention. We explored the associations between parents' and offspring's responses to dynamic facial expressions of emotion, as well as the links between pupil responses and anxiety/depression. Children ($N = 90$, $M_{Age} = 10.13$, range = 7.21–12.94, 47 girls) participated in this lab study with one of their parents (47 mothers). Pupil responses were assessed in a computer task with dynamic happy, angry, fearful, and sad expressions, while participants verbally labeled the emotion displayed on the screen as quickly as possible. Parents and children reported anxiety and depression symptoms in questionnaires. Both parents and children showed stronger pupillary responses to negative versus positive expressions, and children's responses were overall stronger than those of parents. We also found links between the pupil responses of parents and children to negative, especially to angry faces. Child pupil responses were related to their own and their parents' anxiety levels and to their parents' (but not their own) depression. We conclude that child pupils are sensitive to individual differences in parents' pupils and emotional dispositions in community samples.

1 | Introduction

Depression and anxiety are globally the most common forms of psychopathology in youth and adults and are major causes of lifelong global disease burden (GBD 2019 Mental Disorders Collaborators 2022; Polanczyk et al. 2015; Steel et al. 2014; Vos et al. 2020). Depression and anxiety often have an early onset, follow a chronic course, and are highly comorbid (Kessler et al. 2005; Lamers et al. 2011; Scholten et al. 2013; Ter Meulen et al. 2021). Both conditions run in families (Havinga et al. 2017; Micco et al. 2009; Schreier et al. 2006; Telman et al. 2018; Wang et al. 2022). Thus, the presence of depression and/or anxiety in parents constitutes a risk factor for the development of depression and anxiety in their offspring. Both genetic and environmental influences contribute to this risk and to the intergenerational convergence of depression and anxiety.

According to affective science perspectives, depression and anxiety are emotion disorders, characterized by heightened frequency, duration, and intensity of negative emotional experiences, and reduced ability to regulate them (Barlow, Curreri, and Woodard 2021; Gross and Jazaieri 2014). Excessive fear and anxiety states, as well as worry, characterize adult and childhood forms of anxiety disorders. In turn, sadness and irritability often characterize negative affective states in depression (American Psychiatric Association 2013). Cognitive models of depression and anxiety highlight prioritized processing of mood-congruent negative stimuli in the environment as a central mechanism in the development and maintenance of both conditions in youth and adults (Lau and Waters 2017; LeMoult and Gotlib 2019; Platt et al. 2017; Van Bockstaele et al. 2014). According to these models, prioritized processing of threat-relevant stimuli reflects maladaptive information processing in anxious individuals, whereas a broader bias to negative and specifically sad stimuli is characteristic of depression.

A related line of theoretical work highlights the role of parent-to-child transmission of cognitive biases in the intergenerational aggregation of anxiety (Aktar 2022; Creswell et al. 2010). These models propose that the overlap between anxiety in parents and their offspring partly emerges from the intergenerational transmission of anxious biases in the cognitive processing of threat. An anxious processing style is characterized by the prioritized processing of ambiguous, potentially threatening stimuli across the various stages of information processing (e.g., attention and interpretation). The transmission of this anxious processing style is thought to occur via child exposure to parental anxious behaviors and cognitions during everyday encounters of children and parents with ambiguity. According to Field and Lester (2010), each of these encounters constitutes a trial in real-life cognitive bias training. Parents may contribute to the development of cognitive biases in their offspring and thus to the risk of anxiety by displaying a visibly anxious information processing style or engaging in anxious parenting during these encounters. Parent-to-child transmission of cognitive biases, however, has not yet been theoretically addressed in the intergenerational aggregation of depression.

In line with cognitive models of depression and anxiety (Lau and Waters 2017; LeMoult and Gotlib 2019; Platt et al. 2017; Van Bockstaele et al. 2014), anxious children and adults often show an attention bias (AB) toward threat-relevant stimuli (Bar-Haim et al. 2007; Dudeney et al. 2015; Van Bockstaele et al. 2014), whereas depressed youth and adults selectively attend to negative, dysphoric stimuli (Lau and Waters 2017; Peckham et al. 2010; Platt et al. 2017). In contrast, evidence of a significant relationship in AB between parents and their children is scarce or even nonexistent (Aktar et al. 2019; Mogg et al. 2012; Waters et al. 2015, 2018; for a review, see Aktar 2022). The studies addressing overlap in AB between parents and their offspring have thus far relied exclusively on reaction time (RT) measures during computerized tasks, such as the emotional Stroop or visual probe tasks (MacLeod et al. 1986; Williams et al. 1996). In these tasks, AB scores are computed based on the average difference in participants' RTs to threat-relevant versus neutral (or positive) stimuli. In the emotional Stroop tasks, for example, participants are instructed to name the color of threat-relevant versus neutral words (Williams et al. 1996). In the dot-probe task, participants respond to a probe that is presented in the location of either a threat-relevant or a neutral stimulus (MacLeod et al. 1986). RT tasks are known to generate unreliable attentional bias scores in adults, as well as in youth, which may prevent potential parent–child AB relations from emerging (Brown et al. 2014; Schmukle 2005; Sharpe et al. 2022; Van Bockstaele et al. 2020).

In view of the limited reliability of RT measures, alternative eye-tracking measures of attention allocation (i.e., dwell times and fixation duration) have been increasingly incorporated in this line of research (Armstrong and Olatunji 2012; Clauss et al. 2022; Lisk et al. 2020). The relationship between the ABs of parents and their offspring, however, has been investigated using eye-tracking indices of attention only in infancy. This work found a direct link between parent and infant attention to dynamic emotional expressions, as indexed by both fixation durations (Aktar et al. 2022) and pupil responses (Aktar et al. 2021). The current study is the first to study the convergence between parent and offspring emotion processing in childhood using pupillary responses to emotional stimuli as a physiological index of arousal and attention allocation. Under constant luminance, changes in pupil diameter have been argued to reflect activity in the sympathetic nervous system, considered a neural/physiological index of arousal and attention allocation (Bradley et al. 2008; Ferencová et al. 2021). Adult and youth pupils dilate more to positive and/or negative stimuli as compared to neutral stimuli (Bradley et al. 2008; Keil et al. 2018; Price et al. 2012; Silk et al. 2007).

In contrast, valence effects are inconclusive with some studies reporting stronger pupil responses to negative than positive (Kret, Roelofs, et al. 2013; Mckinnon et al. 2020) and other studies finding no significant differences (Bradley et al. 2008; Hepsomali et al. 2017; Kret, Stekelenburg, et al. 2013; Schrammel et al. 2009). These mixed findings may be related to differences in the stimuli used in these empirical studies but also hint at the possibility that maximum levels of arousal are stronger in reaction to negative versus positive emotion displays, indicating potential negativity biases in the perceived intensity. In other words, a stronger response to negative emotions at the apex would suggest dynamic negative emotions are perceived as more intense than positive ones, eliciting stronger arousal in the pupil. The current experiment revisits this comparison in the pupil responses to positive versus negative stimuli using dynamic stimuli. Dynamic stimuli have stronger ecological validity and trigger faster recognition as well as greater accuracy than static emotion displays (Calvo et al. 2016).

Pupils also appear to be sensitive to individual differences in emotional dispositions in adult and child samples. De Zorzi et al. (2021), for example, observed stronger pupil responses to emotional versus neutral stimuli only in highly anxious adults. Hepsomali et al. (2017) reported stronger and slower pupil dilation to emotional and neutral expressions in high-anxious (vs. low-anxious) individuals. Kret et al. (2013) reported stronger pupil responses to angry (but not fearful or happy) facial or bodily expressions in a community sample among participants with higher self-reported trait anxiety. In children, the evidence for an anxiety–pupillary reactivity link is less consistent. Anxious youth were found to show stronger pupil responses when their attention was drawn to the location of fearful versus neutral faces in a study using the dot-probe task in 9-to-13-year-olds (Price et al. 2012). In contrast, there is also some evidence for a weaker, rather than stronger, pupil dilation response to happy, angry, and neutral faces in anxious children: In a study with 10-to-13-year-olds, children diagnosed with social anxiety disorder showed overall reduced pupil responses to faces versus healthy controls

(Keil et al. 2018). Finally, one study reported no significant links between self-reported anxiety levels and pupillary responses to fearful (vs. happy and neutral) expressions in a sample of adolescent girls (12-to-18 years) with subclinical levels of anxiety (Kadosh et al. 2018). Taken together, based on studies that focused on clinical and subclinical samples, there appears to be a link between more anxiety and stronger pupillary reactivity in adults, whereas the evidence for this link is inconclusive in children.

A parallel line of studies addressed the existence of a depression–pupillary reactivity link in adults and youth. For example, adults diagnosed with depression were found to show a stronger sustained pupillary response to both positive and negative words than nondepressed participants (Siegle et al. 2001, 2003). In youth, there is also evidence for a link between depression and pupillary responses, but the precise direction is unclear. For instance, consistent with adult samples, Burkhouse et al. (2017) showed in a sample of 11-to-18-year-olds that children with a lifetime or current diagnosis of depression had stronger pupil responses to positive (happy) and negative (fearful, sad) expressions than healthy controls. Another study, however, revealed a weaker rather than a stronger pupillary response to negative (but not to positive or neutral) stimuli: Silk et al. (2007) reported *less* pupillary reactivity to negative words among currently depressed (vs. nondepressed) 8-to-17-year-olds. Thus, the evidence of a link between depression and stronger pupillary reactivity is relatively scarce and inconclusive, especially for children.

There are also some findings suggesting that child pupillary responses are sensitive to depression/anxiety levels of the parents. For example, a study examining the link between the pupillary reactivity to emotional stimuli in a sample of 8-to-14-year-old children of mothers with (vs. without) lifetime depression and anxiety diagnoses revealed that children of depressed (vs. nondepressed) mothers show stronger pupillary responses to high-intensity sad (but not angry or happy) faces (Burkhouse et al. 2014). In turn, children of mothers with anxiety showed stronger pupillary responses in response to high-intensity angry (but not sad or happy) faces. In additional analyses, researchers found that the differences in child pupillary responses as a function of parental depression/anxiety levels were explained by children's own symptoms in the case of anxiety but not for depression. In the current study, we readdress this issue in a community sample by incorporating parent and child and depression and anxiety symptoms in the models. By studying the links between child emotion processing and emotional dispositions of parents and children, we aimed to bring insights into potential trajectories through which parental and child emotional traits may shape a child's physiological reactivity to others' emotions in typical development.

1.1 | The Current Study

In the studies mentioned earlier, the potential links among depression, anxiety, and pupillary reactivity in adults and children have been mainly tested in samples with elevated (clinical and subclinical levels of) depression and anxiety, whereas it remains currently unknown if these links also hold in community samples. According to developmental psychopathology perspectives, emotion socialization relies on the same mechanisms and processes in adaptive and maladaptive outcomes (Cole, 2016), whereas the links between depression and anxiety with pupillary reactivity and individual differences in pupillary processing accounted for by depression and anxiety levels remain largely unknown in community samples. To our knowledge, the current study is the first to explore these links in a community sample of children and parents.

In this study, we aimed to capture the association between the pupillary responses of parents and their offspring to dynamic facial expressions of emotion and to explore the links between pupillary responses and anxiety/depression in childhood. More specifically, we measured pupillary reactivity in a community sample of 90 children and one of their parents (mother or father) to address 4 questions. First, we investigated the effects of emotion on pupillary responses and explored whether emotion effects on pupil responses differ between parents and children as yoked dyadic pairs (i.e., comparing the magnitude of the pupil response). Given the mixed evidence from earlier studies on the differential pupil responses to negative versus positive emotional expressions (Bradley et al. 2008; Hepsomali et al. 2017; Kret, Roelofs, et al. 2013; Kret, Stekelenburg, et al. 2013; Mckinnon et al. 2020; Schrammel et al. 2009), we explored emotion effects without an a priori hypothesis. In the light of earlier evidence revealing significant differences between adults and children, with stronger pupillary responses to negative versus neutral words in adults than in youth (Shechner et al. 2017), we expected significant differences in parents versus children's pupillary responses.

Second, we examined the direct link between parents' and their offspring's pupillary responses to negative (vs. positive) emotional expressions (i.e., examining the association between the pupil response of the parents and children). Based on earlier evidence in infancy (Aktar et al. 2021), we expected a positive association between parent and child pupillary responses. Third, we investigated the links between pupillary reactivity to negative (vs. positive) emotions and self-reported levels of anxiety and depression in parents and children. Based on earlier evidence indicating a link between depression and anxiety and stronger pupillary responses, especially in adults (Burkhouse et al. 2017; Cascardi et al. 2015; De Zorzi et al. 2021; Hepsomali et al. 2017; Price et al. 2012; Siegle et al. 2003), and no significant differences in the anxiety–pupil link in adults versus youth (Shechner et al. 2017), we expected a positive link between participants' depression and anxiety levels and pupillary responses in both parents and children. Fourth, and finally, we explored the association between parents' depression and anxiety and child pupil responses to negative (vs. positive) emotions. In the light of earlier findings by Burkhouse et al. (2014), we expected specificity in this association such that an increased pupillary reactivity to fearful and angry faces would be apparent in children of parents with higher levels of anxiety, whereas stronger pupillary responses to sad faces would characterize emotion processing in children of parents with higher levels of depression.

2 | Methods

2.1 | Participants and Recruitment

This eye-tracking study was part of a broader research project that focused on emotion processing and parent–child interactions in a community sample of 96 Dutch- or English-speaking parents

TABLE 1 | Sociodemographic characteristics of the parents who contributed to the eye-tracking dataset.

Age *M* (*SD, range*)	43.67 (6.27, 27–72)
Dutch origin % (*Frequency*)	80% (72)
Educational level % (*Frequency*)	
Primary or secondary education	34.44% (31)
Higher professional education	34.44% (31)
Scientific education	24.44% (22)
Other	5.56% (5)
Missing	1.11% (1)
Professional level % (*Frequency*)	
Never worked	2.22% (2)
Mainly manual labor with or without professional training	6.67 % (6)
Mainly brain labor with professional training	11.11% (10)
Independent entrepreneur	21.11% (19)
Salaried at LBO, MBO, or HBO level	46.67% (42)
Salaried employment requiring scientific training	12.22% (11)
Work status	
Unemployed	25.56% (23)
Full-time work	22.22% (20)
Part-time work	18.89% (17)
Student/Scholar	7.78% (7)
Sick leave/Occupational disablement	22.22% (20)
Missing	3.33% (3)

Note: The table summarizes the characteristics of 90 parents who contributed to the eye-tracking dataset.
Abbreviations: HBO = higher professional education, LBO = lower secondary vocational education, *M* = mean, MBO = Dutch secondary vocational education, *N* = sample size, *SD* = standard deviation.

with 8-to-12-year-old children without physical disabilities. All pupil data available from this sample were included in the analyses. For two families, the data were missing because the eye-tracking task was discontinued due to child distress/refusal, whereas data from an additional four families were lost due to experimental errors/calibration failure. Thus, the eye-tracking data were available from 90 families. Children (N = 90, 47 girls [52.2%], M_{Age} = 10.13, SD_{Age} = 1.32, range = 7.21–12.94) visited the lab with one of their parents (N = 90, 47 mothers [52.2%]; for sociodemographic characteristics, see Table 1). The parent who visited the lab with the child was randomly selected. This parent contributed to the eye-tracking data, whereas both parents (if available) were invited to complete online questionnaires (on emotions, depression, anxiety, and emotion regulation). Parents provided written informed consent, and children provided verbal assent for study participation. One family with a child younger than 7.5 was accidentally tested in the study. All analyses were repeated after excluding this child, and the results remain unchanged at $p \leq 0.05$. We therefore decided to keep this family in the final sample to preserve power.

2.2 | Instruments

2.2.1 | Stimuli

We measured children's and parents' pupil responses using dynamic colored emotion stimuli from the Amsterdam Dynamic Facial Expression Set (ADFES; van der Schalk et al. 2011). In ADFES, each dynamic emotional display starts with a 500-ms neutral expression followed by the unfolding of the emotional display, reaching the apex in the next 500 ms, and staying at the apex for another 5000 ms. The stimuli included happy, angry, fearful, and sad videos from two male and two female North European models. The variance stemming from different models was modeled as random effects in all models.

Because our focus was on the differential arousal triggered by positive versus negative expressions and the pupils are sensitive to the changes in luminance and exposure to social stimuli, we have chosen to use the first 500 ms neutral phase of each emotional video as a baseline for luminance and face-exposure (without emotion) effects in the current analysis (also, see Sections 2.4 and 2.5). Neutral trials were not part of this study due to two additional reasons: First, the use of dynamic stimuli may lead to additional error in the measurement of the pupil to non-emotional faces, stemming from the differences in the movement as no muscle movement is visible in the neutral category (only eye blinks), possibly creating a violation of expectations as the rest of the stimuli subsequently turn into emotional displays in all trials. This could have potentially increased cognitive load, confounding the measurement of pupil response. Second, the addition of neutral faces was incompatible with the instruction to name the emotion on the faces in this task (see Section 2.3).

2.2.2 | Questionnaires

Two weeks prior to the lab visit, parents received online links to questionnaires on positive and negative emotions, depression, anxiety, and emotion regulation. Children reported positive and negative emotions, depression, and anxiety during the lab visit with the help of a small dictionary, and the potential help of the experimenters for difficult words. For the current study, our interest was on self-reported depression and anxiety symptoms of parents and children. Children's own reports were chosen to leave out the errors stemming from parental biases in the perception of child depression and anxiety: Parental report of child depression/anxiety has earlier been shown to vary according to parents' own depression/anxiety levels (Kelley et al. 2017). Depression and anxiety scores were available from all 90 parents and children who contributed to the eye-tracking data. Depression and anxiety scores from the parent who completed the laboratory task were used in the analyses.

Depression and anxiety scores were computed as the mean score of all items for all questionnaires for the analyses. In Table 2, we provide descriptive statistics and correlations using sum scores for comparative purposes.

TABLE 2 | Descriptive statistics and correlations between parents' and children's self-reported depression and anxiety scores.

	N	Min.	Max.	*M*	*SD*	1	2	3	4
1. Child depression	90	0	26	7.07	4.66				
2. Child anxiety	90	27	60	40.58	7.68	0.44**			
3. Parental depression	90	0	27	5.82	4.86	0.06	0.28**		
4. Parental anxiety	90	2	38	13.97	7.07	−0.01	0.17	0.64**	

Note: The table provides the raw associations between the self-reported depression and anxiety scores of parents and children. The scores on depression and anxiety are summed and computed in the standard way in this table for comparative purposes.

Abbreviations: M = mean, Max. = maximum, Min. = minimum, N = sample size, SD = standard deviation.

**$p \leq 0.01$ (2-tailed).

2.2.3 | Child Depression

Children reported depression symptoms in the Children's Depression Inventory (CDI, Kovacs 1992), a questionnaire consisting of 27 items scored on a 3-point Likert scale. Cronbach's alpha for the CDI was 0.77 in the current sample.

2.2.4 | Child Anxiety

Children reported on their anxiety symptoms in the short form of the Fear Survey Schedule for Children-Revised (FSSC-R-SF; Muris et al. 2014). The FSSC-R-SF consists of 25 items measuring child fears scored on a 3-point Likert scale. Cronbach's alpha for the FSSC-R-SF was 0.85 in the current sample.

2.2.5 | Parental Depression

Parents filled in the Beck Depression Inventory-II (BDI-II; Beck et al. 1996), a 21-item questionnaire measuring depression symptoms on a 4-point scale. Cronbach's alpha for the BDI was 0.79 in the current sample.

2.2.6 | Parental Anxiety

To measure anxiety symptoms, we asked parents to complete the trait anxiety subscale of State-Trait Anxiety Inventory (STAI; Spielberger et al. 1983). This subscale consists of 20 items scored on a 4-point scale. Anxiety scores were obtained by averaging parents' responses to these items. Cronbach's alpha for the trait anxiety subscale was 0.86 in the current sample.

2.3 | Procedure

Pupil responses of parents and children were measured during the presentation of emotional expressions (1280 × 1024 pixels) using an EyeLink eye-tracker (500 Hz). The task consisted of four emotional displays from four models presented in random order in three successive blocks, resulting in 48 trials. Each trial started with a 500 ms attention getter, followed by a 1000 ms blank screen and a 1500 ms blurred facial display, and ended with the dynamic emotional stimulus (starting with a neutral expression for 500 ms, followed by the unfolding of emotion till apex for 500 ms, and ending with the emotion at the apex for another 5500 ms) (see Section 2.2.1 and Figure 1a for the time flow of a trial).

Before the start of the task, children and parents were informed about the experiment. Parents and children received the instruction to verbally name the emotion displayed on the screen in each trial as fast as possible. This aimed to facilitate child engagement in the experiment. Emotion recognition corresponds to a later stage of information processing than pupillary responses, which remain unaffected by the instruction to name the emotion as compared to the free viewing format (Snowden et al. 2016). As such, we did not incorporate emotion recognition measures for the current study. An external twin camera was used to record participants' responses. Parents and children could not see each other while performing the task.

Participants were seated approximately 60 cm away from the screen. The gaze was calibrated and validated with 5-point procedure covering the four corners and the center of the screen. Parents completed the task after the children. A random stimulus order was generated for children and identically repeated for the parent. Following the eye-tracking task, parents and children were invited to a different lab for interaction tasks (not used in the current study), and child questionnaires were also completed in this lab.

2.4 | Data Reduction

The current study focused on the pupillary responses of parents and children to emotional expressions as a physiological index of arousal. Parents' and children's pupillary responses to emotional expressions were extracted using EyeLink Data Viewer (SR Research Ltd., Mississauga, Canada). The data from parents and children were processed and analyzed in a uniform way to allow for a direct comparison.

First, outlying values (>|3| *SD*s) and sudden jumps between two consecutive time points were removed from the individual distribution of pupil responses. Next, missing observations were replaced using linear interpolation. Following the interpolation, the pupil responses were aggregated to 60 observation points with 100 ms intervals during the presentation of dynamic stimuli. To control for luminance and face exposure effects in the absence of emotional expressions, pupil responses were baselined (via division) to the mean pupil responses during the first 500 ms of the video, where the dynamic expression displayed no emotion/was

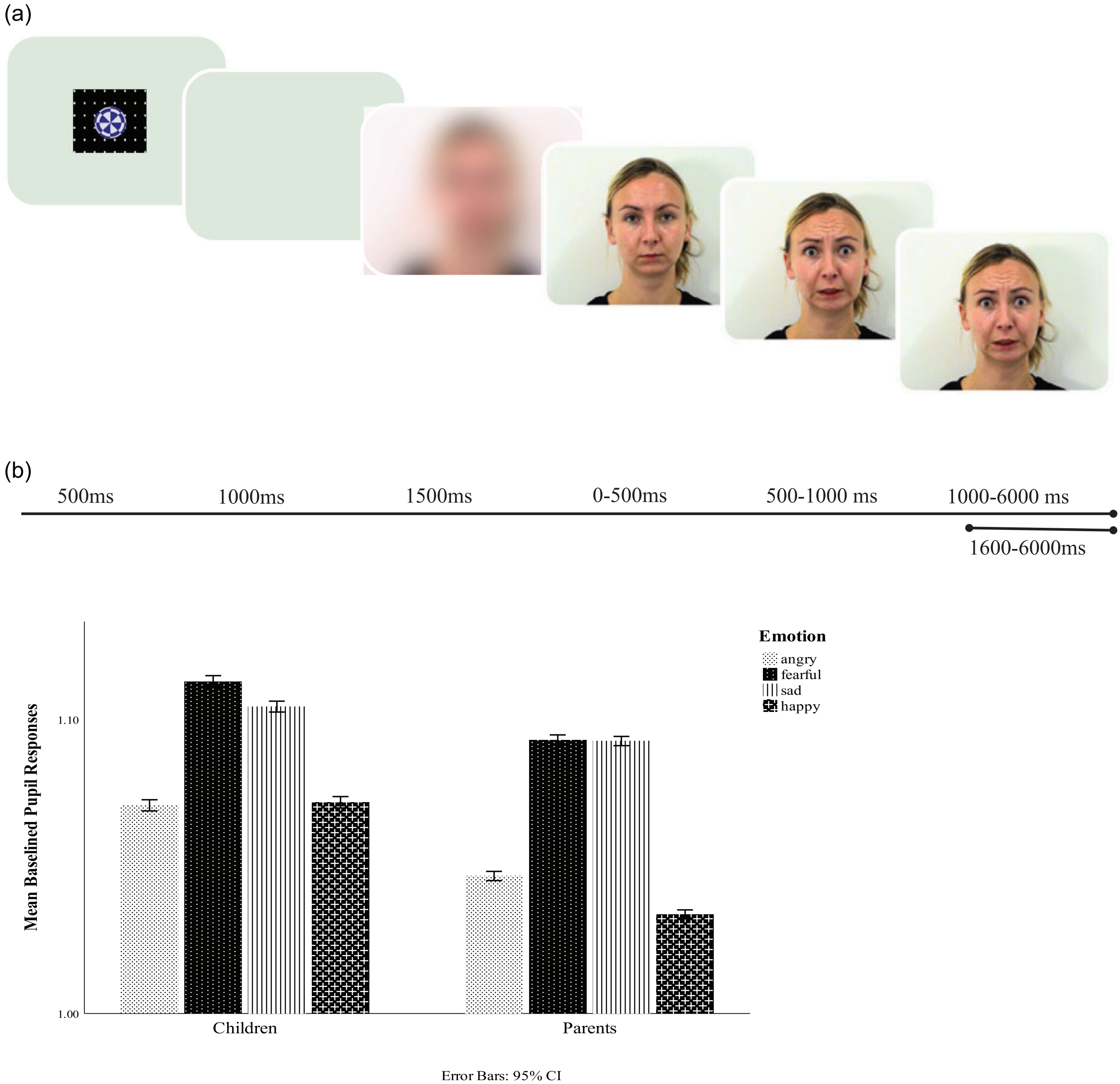

FIGURE 1 | (a) Time flow of a trial in the current experiment and (b) mean pupil responses of children and parents to negative versus positive facial expressions.

neutral. Thus, the pupil responses in the current study are a measure of the relative change in pupil size from the neutral baseline. Pupil responses to emotional faces were analyzed following the light pupillary reflex (between 1500 and 6000 ms; thus, 500 ms after the expression reaches the apex). Outlying values in the distribution of baseline pupil responses (>|3| *SD*s) were removed. Trials, in which the data were available for 1 s or longer within the 6 s presentation time (missing < 83.33%), were included in the analyses. The mean percentage of missing observations was 5.7% (*SD* = 11.74, range: 0–82.8) in children and 4.8% (*SD* = 11.08, range: 0–82.7) in parents. Participants who had data in less than 10 trials were excluded from the dataset (data from one parent removed at this stage). The pupil data were available on average for 47.63 trials (*SD* = 2.12, range: 29–48) from children and 47.59 trials (*SD* = 2.65, range: 24–48) from parents.

2.5 | Statistical Analyses

This dataset has a repeated hierarchical structure with 45 observation points per trial, nested over 48 trials, and nested over family members (parent and child). To account for the hierarchical structure, the research questions were analyzed using mixed models (repeated multi-level regressions) that contained the family at the highest level, followed by family member, and repeated observations across and within trials at the lowest level. An autoregressive covariance structure was used to account for the repeated observations of pupil over stimulus time. Baselined pupil responses were the main outcome measure in all models. Thus, the present results concerning pupil responses should be understood as a change in pupil size relative to a neutral baseline. Maximum likelihood was the estimation method. The intercept, as well as Model ID and Trial number (TrialID), were randomized in the models. The categorical predictors of emotion (angry, fearful, sad vs. happy) and family member (child vs. parent), as well as the continuous predictors on depression/anxiety scores, were entered as fixed effects in all models. For all comparisons involving emotion and family member, the happy face and parents were the reference, respectively. Depression and anxiety scores from the parent who completed the laboratory task were used in the analyses. The distribution of depression and anxiety scores showed sufficient normality (skewness and kurtosis were $\leq |2.5|$), and the scores were standardized in the analyses.

Descriptive statistics are presented in Table 2. Mean BDI scores in parents correspond to the 40th and 60th percentile range of the distribution reported for highly educated Dutch community samples, suggesting good generalizability to typical community samples (Roelofs et al. 2013). Mean depression scores were below the clinical cut-off for parents (14 and 18; Roelofs et al. 2013) and for children (16, Emons et al. 2019), suggesting low-to-moderate levels of depression in the current sample. In contrast, the anxiety symptoms were overall closer to the clinical threshold, especially for children. For FSSC-R-SF, which we used to measure child anxiety, the cut-off scores between 45 and 52 correspond to the 90th percentile in Dutch community samples (Muris et al. 2014), with current means corresponding to moderate, subclinical levels of anxiety in children. The mean levels of parental anxiety in the current sample remain below the clinical cut-off of 40 in STAI, pointing to low/moderate anxiety levels in the current sample.

The raw associations between depression and anxiety scores of parents and children are also presented in Table 2. There were moderate positive associations between self-reported depression and anxiety, both for parents and children. There was also a significant correlation between child anxiety and parental depression, whereas the correlation between parents' and children's depression and between parents and children's anxiety, $p = 0.116$, was not significant. Mothers and fathers did not significantly differ in their pupil responses ($p = 0.629$), but did differ in self-reported depression t (85.22) = 2.21, $p = 0.030$, and anxiety levels t (87.87) = 2.93, $p = 0.004$. Mothers reported more depression ($M = 6.87$, $SD = 5.33$) and more anxiety ($M = 15.96$, $SD = 7.19$) than fathers ($M = 4.68$, $SD = 4.05$ and $M = 11.79$, $SD = 6.32$ for depression and anxiety, respectively).

Given the significant correlations between within-person anxiety and depression, we chose to analyze the results in two successive models, one containing child and parent depression and the other containing child and parental anxiety. The initial models for each research question included the main effects and the theoretically relevant interactions of interest. We have reached the final models by removing the nonsignificant interactions from the model one-by-one, starting with the estimates with bigger p-values.

3 | Results

3.1 | Do Pupil Responses to Negative (vs. Positive) Facial Expressions Differ Between Children and Parents?

The model comparing pupil responses of children and parents to negative versus positive emotions included the main effect of emotion (angry, fearful, sad vs. happy), family member (parent vs. child), and the two-way interaction between emotion and family member. The interaction was significant, $N = 180$, F (3, 164,626.42) = 68.86, $p < 0.001$, and was retained in the model (presented in Table S1). Thus, parents and children differed in the intensity of their pupil responses to negative versus positive facial expressions. Children showed stronger differential pupil responses to emotional expressions than parents to angry ($\beta = 0.06$ $SE = 0.02$, $p = 0.001$), fearful ($\beta = 0.14$, $SE = 0.02$, $p < 0.001$), and sad ($\beta = 0.24$, $SE = 0.02$, $p < 0.001$, see Table S1) faces. A quick comparison of mean baselined pupil responses of parents and children across emotions, presented in Figure 1b, reveals that children had overall stronger pupil responses to all emotional expressions than parents.

Next, the emotion effects were tested in two separate models for children and parents. The effect of emotion was significant both on children's, $N = 90$, F (3, 7640.70) = 44.03, $p < 0.001$, and parents' pupil responses, F (3, 7860.23) = 115.50, $p < 0.001$, indicating that both children and parents had different pupil responses to negative versus positive faces. Children showed stronger pupil responses to fearful ($\beta = 0.19$, $SE = 0.02$, $p < 0.001$) and sad ($\beta = 0.16$, $SE = 0.02$, $p < 0.001$) than happy faces, whereas child pupil responses did not differ between angry and happy faces, $p = 0.914$. In turn, parents showed significantly stronger pupil responses to all negative emotions as compared to happy: angry ($\beta = 0.05$, $SE = 0.02$, $p = 0.010$), fearful ($\beta = 0.31$, $SE = 0.02$, $p < 0.001$), and sad ($\beta = 0.30$, $SE = 0.02$, $p < 0.001$).

To summarize, we found differences in how parents' and children's pupils responded to negative versus positive emotions. Child baselined pupil responses to emotional expressions were overall stronger than parents' pupil responses. Separate analyses for parents and children revealed that parents responded to all negative faces with increased pupil responses relative to happy faces, whereas children only showed enhanced pupil responses only to fearful and sad but not to angry faces, as compared to happy faces.

3.2 | Is There a Direct Relationship Between Parents' and Their Offspring's Pupil Responses to Negative (vs. Positive) Emotional Expressions?

To capture the direct link between parents' and children's pupil responses, we ran multi-level models with child pupil responses as the outcome and parents' pupil responses as the predictor. This model ($N = 90$) also included the main effect of emotion, and the two-way interaction between emotion and parents' pupillary responses. This interaction was significant, F

(3, 173,421.38) = 4.79, $p = 0.002$, and was retained in the model (presented in Table S2). Thus, the link between parent and child pupil responses differed as a function of emotion. The strength of the direct link between parent and child pupil responses did not differ between angry versus happy facial expressions, $p = 0.123$, whereas this link was less pronounced for fearful, $\beta = -0.02$, $SE = 0.01$, $p = 0.011$, and for sad than happy expressions, $\beta = -0.03$, $SE = 0.01$, $p < 0.001$. Thus, the convergence between parent and child differential pupil responses was significant, especially for happy and angry expressions.

3.3 | Are Depression and Anxiety Related to Pupil Responses to Negative (vs. Positive) Emotions in Parents and Children?

The model comparing the links between pupil responses to negative emotions and *depression* levels in parents and children included the main effects of emotion (angry, fearful, sad vs. happy), family member (parent vs. child), and depression, and all the two- and three-way interactions between these effects. The three-way interaction was significant and was retained in this model, $N = 180$, F (3, 78,347.30) = 2.68, $p = 0.046$.

Parallel models comparing the links between pupil responses and anxiety levels in parents and children also revealed a significant three-way interaction between emotion, family member, and anxiety, $N = 180$, F (3, 91,460.92) = 8.20, $p < 0.001$. Thus, the associations between pupillary responses to emotion and both depression and anxiety differed significantly between parents and children and between negative versus positive emotions. We therefore analyzed these links separately in parent and child models.

3.3.1 | Are Depression and Anxiety Related to Pupillary Responses to Negative (vs. Positive) Emotions in Parents?

We investigated the association between parents' pupil responses to negative versus positive emotions and depression/anxiety in multi-level models with parents' pupil responses as the outcome, and depression/anxiety as the predictor, in addition to the main effects of emotion and the interaction between emotion and depression/anxiety. In the depression model (presented in Table S3), there was a significant interaction between emotion and depression, $N = 90$, F (3, 7864.76) = 2.80, $p = 0.039$. Standardized estimates revealed a significant difference in the link between parental depression and pupil responses to fearful versus happy expressions ($\beta = -0.05$, $SE = 0.02$, $p = 0.017$) and to sad versus happy expressions ($\beta = -0.05$, $SE = 0.02$, $p = 0.016$), whereas this link did not differ between angry versus happy expressions, $p = 0.383$. The plot of this interaction, presented in Figure 2a, revealed a positive slope for the link between parents' depression and their pupil responses for happy faces, which did not differ significantly from the slope of the association for angry faces, whereas the link between parents' depression and their pupil responses for sad as well as fearful faces had a negative slope. Thus, the findings suggest that parents reporting higher levels of depression showed stronger pupil responses to happy and angry, and a reduced response to fearful and sad facial expressions.

In the anxiety model (presented in Table S4), there was a significant interaction between emotion and anxiety, F (3, 7884.72) = 3.64, $p = 0.012$. Standardized estimates revealed a significant difference in the link between parental anxiety and pupil responses to angry versus happy expressions, $\beta = -0.04$, $SE = 0.02$, $p = 0.040$, and between parental anxiety and pupil responses to sad versus happy expressions, $\beta = -0.07$, $SE = 0.02$, $p = 0.001$, whereas this link did not differ between fearful versus happy expressions, $p = 0.210$. The plot of this interaction, presented in Figure 2b, reveals overall positive links between parents' pupil responses and anxiety. The slope of this link between parents' self-reported anxiety and differential pupil responses was bigger for fearful (which did not significantly differ from happy faces), as compared to angry and sad faces. In sum, the positive link between parents' anxiety and pupillary responses was especially pronounced for happy and fearful faces.

3.3.2 | Are Depression and Anxiety Related to Pupil Responses to Negative (vs. Positive) Emotions in Children?

We investigated the association between child pupil responses to negative versus positive emotions and child depression in an initial multi-level model with child pupil responses as the outcome and child depression as the predictor, in addition to the main effect of emotion and the interaction between depression and emotion. The interaction was not significant, $p = 0.145$, and was removed from the model, reducing the final model to the main effects of emotion and depression. The final model did not reveal a significant association between depression and pupil responses, $p = 0.345$.

Next, we ran this model with child anxiety as a predictor. The interaction was significant, F (3, 7651.49) = 6.83, $p < 0.001$, and was retained in the final model (presented in Table S5). The standardized estimates revealed no significant differences in the link between child anxiety and pupil responses for angry versus happy, $p = 0.424$, and fearful versus happy, $p = 0.640$, expressions, whereas this link differed between sad versus happy $\beta = -0.08$, $SE = 0.02$, $p = 0.001$ expressions. The plot of this interaction, shown in Figure 3, reveals a negative link between child anxiety and pupil responses to sad faces, whereas a positive slope was observed between child anxiety and pupil responses to happy, angry, and fearful faces. In other words, more anxious children responded with reduced pupillary arousal to sad faces and enhanced pupillary arousal to happy, angry, and fearful faces.

3.4 | Are Parents' Depression and Anxiety Related to Child Pupil Responses to Negative (vs. Positive) Emotions (After Taking Child Depression/Anxiety Levels Into Account)?

To investigate the association between child pupil responses to negative versus positive emotions and parent depression/anxiety, the main effects of parental depression/anxiety as well as the two-way interaction between emotion and parental depression and between emotion and parental anxiety were added as additional predictors to the models presented above in Section 3.

In the multi-level regression of child pupillary responses on parental and child depression, the interaction between emotion

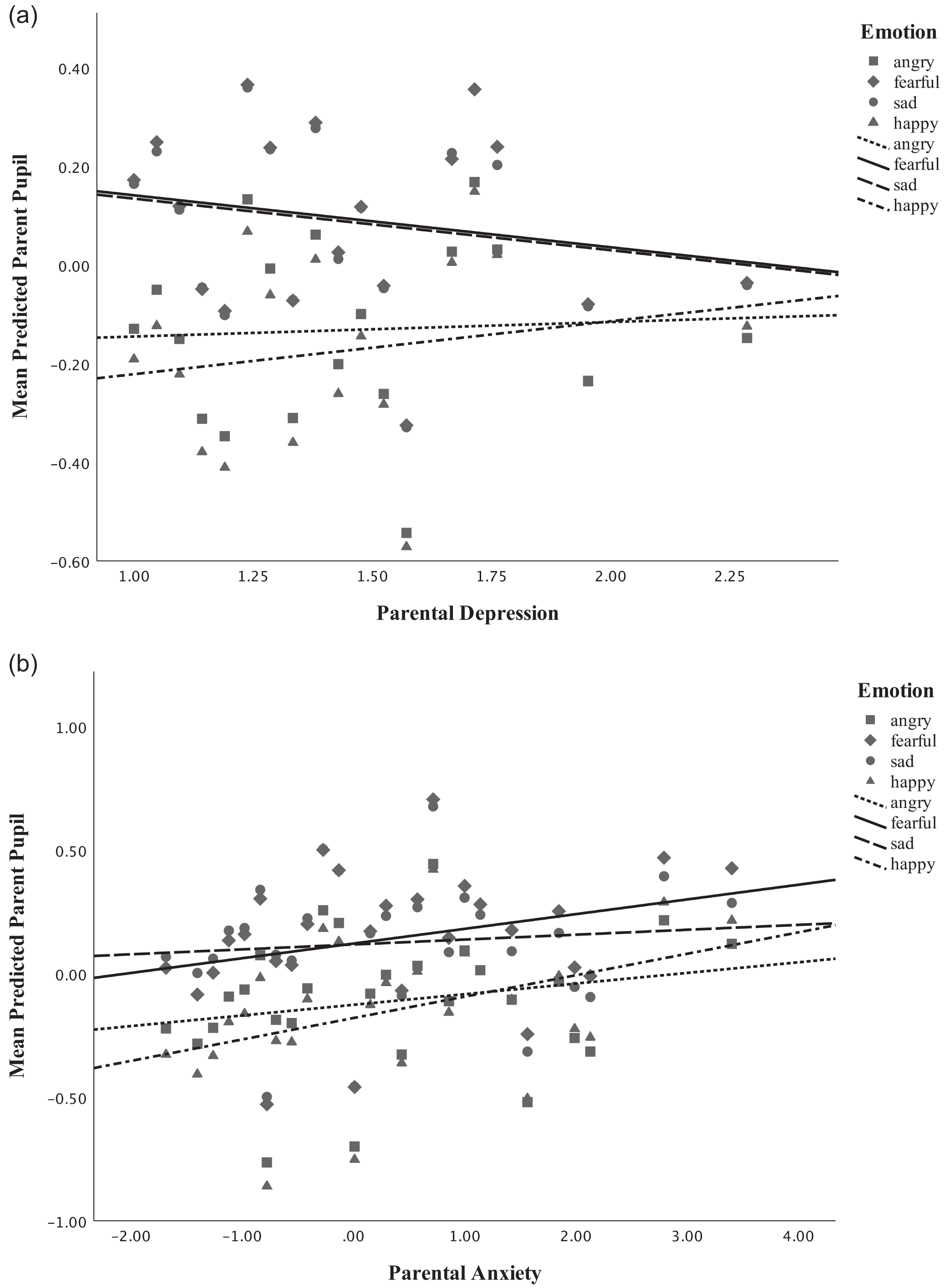

FIGURE 2 | The scatterplot of the links between (a) predicted mean pupil responses and depression and (b) predicted mean pupil responses and anxiety in parents.

and parent depression was significant, F (3, 7643.55) = 7.06, $p < 0.001$, and retained in the final model presented in Table S6. The link between parental depression and child pupil responses did not differ between angry versus happy, $p = 0.099$, and between sad versus happy faces, $p = 0.241$, whereas the link was stronger for fearful faces as compared to happy faces, $\beta = 0.06$, $SE = 0.02$, $p = 0.011$. The plot of this interaction is presented in Figure 4a. The slope of the link between parental depression and child pupil responses was positive for fearful faces, whereas it was negative for happy, angry, and sad faces. In other words, children of parents who reported higher levels of depression showed enhanced pupillary arousal to fearful faces, whereas a reverse trend was observed with reduced pupillary arousal to happy, angry, and sad faces.

In the anxiety model, the interaction between emotion and parental anxiety was significant, F (3, 7643.33) = 8.25, $p < 0.001$, and was retained in the model presented in Table S7. Standardized estimates revealed a significant difference in the link between parental anxiety and child pupil responses to fearful versus happy

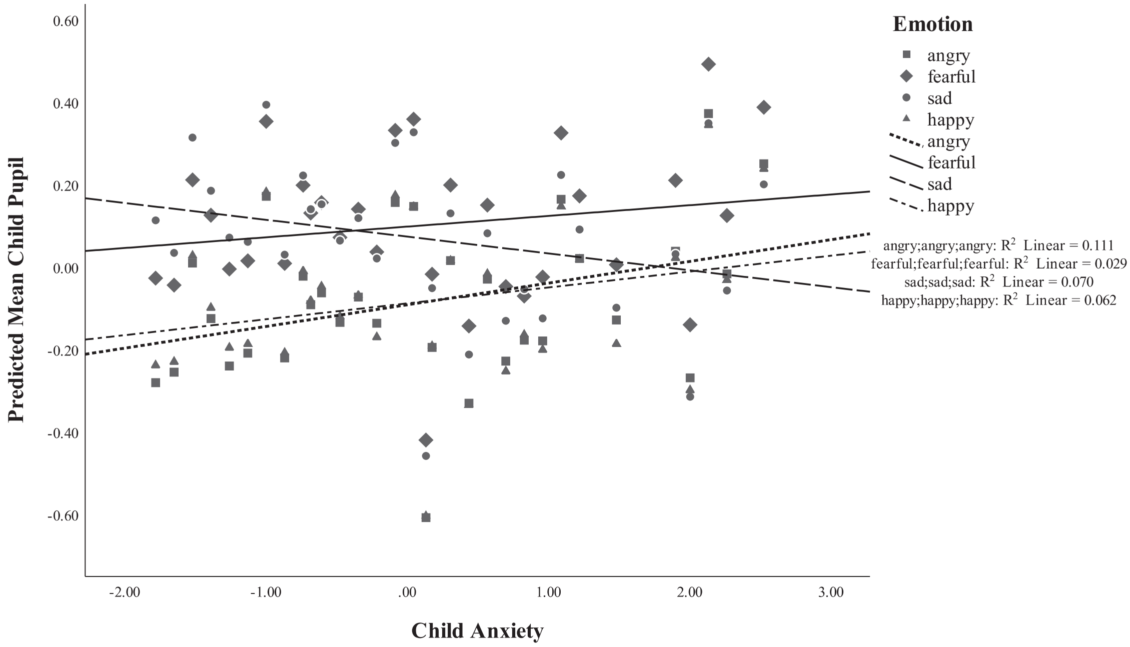

FIGURE 3 | The scatterplot of the links between predicted mean pupil responses and anxiety in children.

expressions, $\beta = 0.08$, $SE = 0.02$, $p < 0.001$, whereas this link did not differ between angry and happy, $p = 0.234$, or between sad and happy faces, $p = 0.215$. The plot of this interaction, presented in Figure 4b, reveals that the slope of the link between parents' anxiety and child pupil responses was positive for fearful faces, whereas it was negative for the link between parents' anxiety and child pupil responses to happy, angry, and sad facial expressions. Thus, children of parents with more anxiety show enhanced pupillary arousal to fearful faces and reduced pupillary arousal to happy, angry, and sad faces. The interaction between emotion and child anxiety remained significant in this model that incorporated parental anxiety as an additional predictor, F (3, 7661.32) = 7.72, $p < 0.001$, see Table S7).

4 | Discussion

This study was the first to examine pupillary responses of parents and their 8-to-12-year-old children to dynamic emotional facial expressions as an index of physiological arousal and attention allocation. We explored the similarities and differences between parent and child pupillary responses and the link between pupillary responses and self-reported anxiety and depression levels in parents and their offspring. Below, we discuss our findings in more detail per research question.

4.1 | Do Pupil Responses to Negative (vs. Positive) Facial Expressions Differ Between Children and Parents?

Parents and children differed in their pupillary responses to negative versus positive emotions. We found overall stronger differential pupillary responses in children, as compared to parents. Stronger pupillary responding in children may indicate that the regulatory skills for emotional arousal are still developing in childhood years. This finding is somewhat inconsistent with earlier evidence revealing differences indicating overall stronger differential pupillary responses in adults than in youth to negative versus neutral words (Shechner et al. 2017). This inconsistency may be related to a more pronounced pupil response to neutral than positive stimuli in children due to the inherent ambiguity of the dynamic neutral face (earlier reported in infancy by Aktar et al. (2021)) in this earlier study. Alternatively, it can be explained by the differential arousal levels evoked by the dynamic emotional faces compared to emotional words, especially considering the differences in reading skills between children and adults that would not manifest in the processing of emotional faces.

Separate analyses of emotion effects in parents and children revealed that parents responded to all negative versus positive emotions with increased pupillary arousal. In contrast, children showed an enhanced pupillary response to fearful and sad but not to angry, as compared to happy faces. Interestingly, a relatively attenuated pupillary response to angry faces characterized the emotional processing of both parents and children. To our knowledge, our study was the first to incorporate angry faces. Further studies will be needed to explain why dynamic anger stimuli that signal direct threat would create a relatively inhibited pupillary response, thus less arousal than other negative emotions.

4.2 | Is There a Direct Relationship Between Parents and Their Offspring's Pupil Responses to Negative (vs. Positive) Emotional Expressions?

In line with the earlier evidence revealing a significant convergence between parents and their offspring's pupillary responses in infancy (Aktar et al. 2021), we found a relationship between pupillary responses between parents and the offspring. Thus, despite a relatively reduced differential response to angry faces in children, there was an overall synchrony between parents'

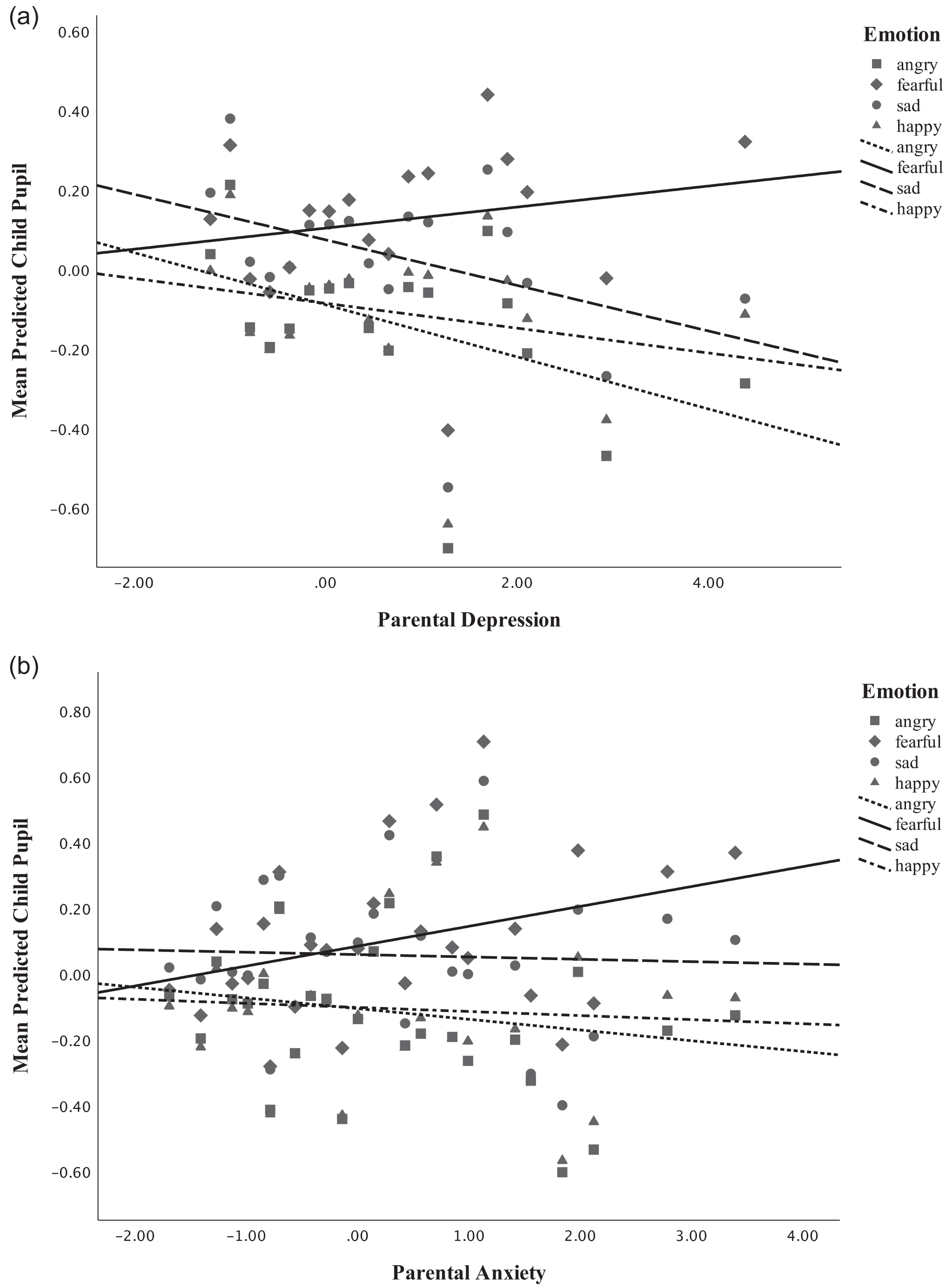

FIGURE 4 | The scatterplot of the links between (a) mean predicted child pupil responses and parental depression and (b) mean predicted child pupil responses and parental anxiety.

and children's pupillary arousal in response to emotional faces. Interestingly, the magnitude of this synchrony was especially strong for angry and happy compared to fearful or sad faces. This may be related to angry and happy affective states being relatively more common in daily parent–child interactions, which could mean that they are more often co-regulated and therefore better synchronized between parents and children. To our knowledge, this is the first study capturing the parent–child convergence in pupillary indices of attention allocation in a community sample. Earlier evidence using notably unreliable RT measures to measure attentional bias (AB) may have failed to capture such emotion-specific convergence of parent–child AB (Aktar et al. 2019; Mogg et al. 2012; Waters et al. 2015, 2018; for a review, see Aktar 2022) because of their low reliability. Pupillary responses may provide a relatively more powerful alternative physiological index of attentional bias due to their higher sampling rate and temporal sensitivity. This convergence in parents' and children's physiological responses to emotion is likely to be driven by separate and joint influences of genetic factors and environmental processes. Indeed, they may be driven by the same mechanisms/processes that account for the intergenerational convergence in emotional disorders.

4.3 | Are Depression and Anxiety Related to Pupillary Responses to Negative (vs. Positive) Emotions in Parents and Children?

The current findings suggest that the link between pupillary responses and depression/anxiety differs between parents and children. For depression, we noted a positive link between self-reported depression and pupillary responses to happy and angry faces only in parents. With respect to anxiety, there was a link between more anxiety and stronger pupillary responses to negative emotions for both parents and children. However, the anxiety–pupil link between parents and children was different during sad faces: Parents with higher levels of self-reported anxiety responded with stronger pupil responses to sad faces, whereas children with higher levels of self-reported anxiety responded with weaker pupil responses to sad faces. With this exception for sad faces, the anxiety–pupil link was somewhat similar across parents and children, which is more consistent with the earlier evidence that revealed no significant differences between adults and children in the link between pupillary reactivity to negative words and anxiety (Shechner et al. 2017).

The link between more anxiety and stronger pupil responses in parents is in line with earlier evidence suggesting a link between higher levels of anxiety and pupillary vigilance to positive and negative emotional facial expressions in adults (De Zorzi et al. 2021; Hepsomali et al. 2017). Interestingly, in the current study, the links of self-reported anxiety to physiological arousal were especially strong during happy and fearful faces. Thus, this enhanced reactivity does not appear to be specific to negative or threat-relevant emotions as it extended to positive and non-threat-relevant negative emotions.

We also found stronger pupillary responding in parents with higher levels of self-reported depression, but only to angry and happy faces. In contrast, a reduced pupil response to fearful and sad facial expressions was observed in parents who reported higher levels of depression. Thus, our findings are only partially consistent with earlier evidence that revealed a link between depression diagnosis and a vigilant pupillary response to negative word stimuli in adults (Siegle et al. 2001, 2003). Our results further show that this vigilance was only visible for expressions that directly signaled the active presence or absence of threat (anger and happiness), whereas there was a mood-congruent dampening for fear and sadness.

Taken together, the findings provide support for the idea that pupillary responses to emotional expressions are sensitive to individual differences in emotional dispositions in a community sample of adults. Pupillary responses showed a predominantly vigilant pattern of responding that was not specific to negative or threat-relevant expressions in parents with higher levels of depression or anxiety. This vigilant pattern may be indicative of an enhanced overall readiness of the nervous system to respond to emotional stimuli in individuals who report more depression or anxiety. The reduced responding to mood-congruent, fearful, and sad faces in relation to depression may be indicative of an avoidant or dampened responsivity.

In children, we found no relations between pupillary responding and depression. Thus, this link may be characteristic of clinical rather than community samples of depressed children, where the overall levels of depression remain in low-to-moderate levels (Burkhouse et al. 2014). In contrast, we did capture the link between anxiety levels and pupil responses in the current community sample: More anxious children responded with reduced pupillary arousal to sad faces and enhanced pupillary arousal to happy, angry, and fearful faces. This latter pattern is partly in line with earlier findings revealing a link between anxiety and vigilant pupillary responses to threat-relevant stimuli in children (Price et al. 2012). Here, higher levels of self-reported anxiety in children were related to more vigilant pupillary responding not only to threat-relevant negative but also to happy facial expressions, whereas higher levels of anxiety were linked to a diminished pupillary responding to sad faces, which may be related to the fact that sad faces do not (directly or indirectly) signal threat. The lack of specificity to positive or threat-relevant expressions is in line with early pupil findings showing an overall enhanced pupillary reactivity to emotion in children from clinical samples (Burkhouse et al. 2017).

4.4 | Are Parents' Depression and Anxiety Related to Child Pupil Responses to Negative (vs. Positive) Emotions (After Taking Child Depression/Anxiety Levels Into Account)?

We noted significant links between child pupillary responses and parents' depression/anxiety in the models that also included children's own depression/anxiety levels as predictors. In contrast with earlier findings from a clinical sample (Burkhouse et al. 2014), no specific vigilant responding to sad faces in children of parents who reported higher levels of depression was observed. Our findings point in the opposite direction. We found that children of parents who reported higher levels of depression had a reduced pupillary responding, not only specifically to sad but also to angry and happy faces, whereas they showed a vigilant pupillary response only to fearful faces. The differences in findings of the current and this earlier study may be related to the differences in the samples (clinical vs. nonclinical) or stimuli (morphed static vs. dynamic).

A similar pattern was observed in the model with parental anxiety, with a significant interplay between emotion and parental anxiety. The link between parental anxiety and child pupillary responses depended on emotion. Children of parents with higher levels of self-reported anxiety showed more reduced pupillary responses to happy, angry, and sad faces and more vigilant pupillary responses to fearful faces. The findings are roughly in line with earlier evidence demonstrating vigilant pupillary responding to threat-relevant expressions in children of anxious parents (Burkhouse et al. 2014), except that in our study, vigilant responding was only present while processing fearful but not angry faces. In fact, more anxiety in parents was related to *reduced* pupillary arousal to angry expressions, which may be related to the direct signal value of angry faces.

5 | Limitations and Future Directions

Our study also had the following limitations. First, our correlational design does not allow any inferences on causality or on the timeline of the reported associations, including the one between parent and child pupillary responses. Future studies exploring this link in genetically sensitive longitudinal designs that include community and clinical samples will be needed to establish the developmental timeline of these associations and to delineate genetic and environmental mechanisms explaining them. Second, the current dataset on parent and offspring pupil responses only included one parent from each family, and the sample size was too small to investigate interparental differences in the link between child pupil responses and parents' pupil responses, depression, and anxiety. Future studies should consider including both parents' pupil responses to allow a comparison of the links with mothers versus fathers. Third, our experimental tasks included the instruction to label the emotion on screen. Although one may argue that the instruction to label the emotion may have interfered with the measurement of pupillary responses, earlier evidence suggests that pupillary responses are not necessarily influenced by naming emotional stimuli (Snowden et al. 2016). In the current study, we chose to exclusively focus on physiological responses, at the expense of leaving out accuracy and RT measures, in order to explore a more powerful and temporally sensitive alternative physiological index of attentional bias. Fourth, although the current study has sufficient power to analyze the main effects of emotion in parents and children (accumulated from the repeated observations of pupil dilation over time in the current multi-level models), the relatively small sample size may potentially lead to this study being underpowered for the analysis of individual differences in pupillary responses. Testing these associations in large community samples is an essential next step, especially because the effect sizes reported for the link between parents' depression/anxiety and child people were of small to medium magnitude in clinical samples (Burkhouse et al. 2014), which may be even smaller for community samples. Finally, the current sample consisted of highly educated parents, and the findings may not be generalizable to parents with lower Socioeconomic status (SES).

Despite these limitations, the current study was the first to capture the direct link between parent and offspring processing of others' emotions using pupillary indices and among the first to showcase the links between these pupillary responses and self-reported anxiety/depression symptoms in a community sample of parents and children. The observed parent–offspring convergence in physiological indices of emotion processing in community samples is likely to be driven by the same mechanisms that explain the intergenerational convergence in emotion-related psychopathology in clinical samples.

Acknowledgments

The authors are grateful to the parents and children who participated in this study and to Eline Möller, Stephanie Mizrahi, and the members of Amsterdam Family Lab for their support during the realization of this project.

Ethics Statement

This study (2015-CDE-4037) was approved by the Ethics Review Board of the Faculty of Social and Behavioral Sciences, University of Amsterdam, on February 17, 2015.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All raw data and materials will be stored online within 1 month after publication in DataVerse at https://dataverse.nl/, available upon request from the corresponding author.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.